

Project 1: Python FastAPI Weather Proxy CI/CD Demo

Overview

This project demonstrates a complete CI/CD pipeline using Jenkins, SonarQube, Docker, and Trivy for a Python FastAPI application.

Prerequisites

Ensure the following are installed and running on your Jenkins server:

- Docker
- Trivy
- Python 3 & pip

Step 1: Network Setup

Create a shared Docker network for your infrastructure and deployments:

```
docker network create jenkins-net
```

Step 2: Infrastructure Setup

Run SonarQube and Jenkins on the `jenkins-net` network:

```
# Run SonarQube
```

```
docker run -d --name sonarqube -p 9000:9000 --network jenkins-net  
sonarqube:lts-community
```

```
# Run Jenkins (with Docker socket mounted)
```

```
docker run -d --name jenkins -p 8080:8080 -p 50000:50000 --network  
jenkins-net -v /var/run/docker.sock:/var/run/docker.sock -v  
jenkins_home:/var/jenkins_home jenkins/jenkins:lts
```

Step 3: Jenkins Configuration

1. Log into Jenkins at `http://localhost:8080`.
2. Install necessary plugins: Docker Pipeline, SonarQube Scanner, Pipeline.
3. Configure SonarQube Server:
 - Go to Manage Jenkins -> System.
 - Add a SonarQube server named SonarQube and point it to http://sonarqube:9000.
4. Configure Global Tools:
 - Go to Manage Jenkins -> Tools.

- Add a SonarQube Scanner installation.

Step 4: Pipeline Execution

1. Create a new Pipeline job in Jenkins.
2. Under the Pipeline section, choose "Pipeline script from SCM" (if hosted on Git) or paste the contents of Jenkinsfile.
3. Run the build.

Step 5: Verification

Once the pipeline finishes successfully, access the deployed FastAPI app:

- URL: `http://localhost:8000`
- Expected Output: `{"Hello": "Weather Proxy"}`